Firewalls in Computer Networks

Protecting Networks from Unauthorised Access

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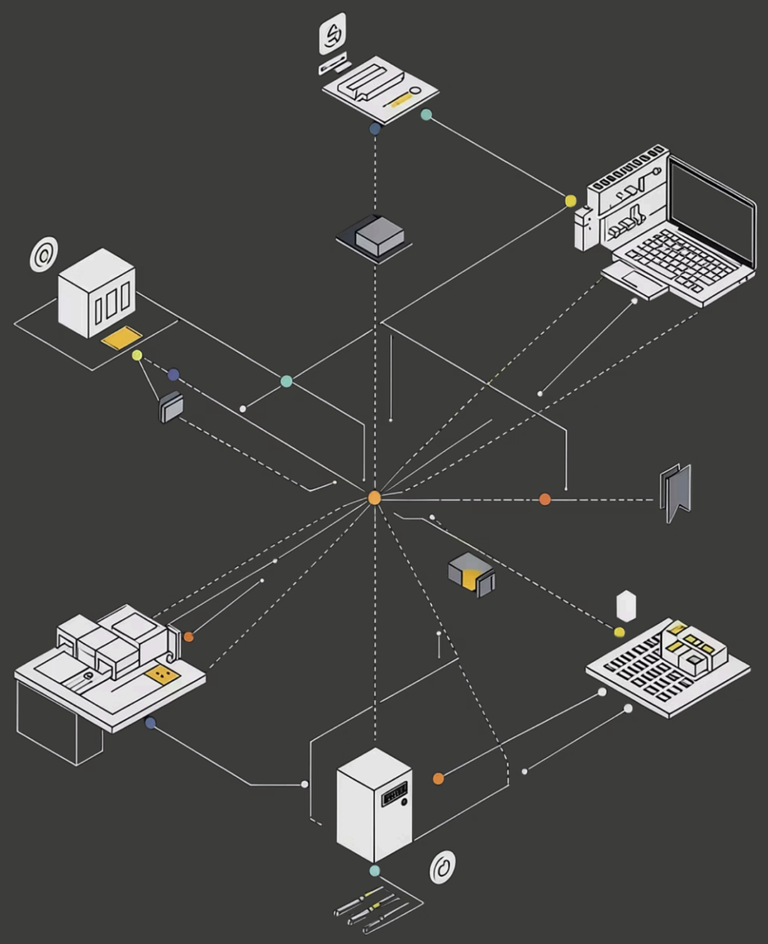
What is a Firewall?

Definition

A network security device that monitors and filters incoming and outgoing network traffic based on predetermined security rules.

Real-life Analogy

Think of a firewall as a security guard at a gate who checks every person entering or leaving, allowing only authorised individuals through.



How Firewall Works

1

Monitor Traffic

Watches all data packets moving in and out of the network

2

Apply Rules

Checks each packet against security rules and policies

3

Allow or Block

Decides whether to permit or deny each connection attempt

Types of Firewalls

Hardware Firewall

Physical device protecting entire networks. Used by organisations for centralised security.

Software Firewall

Installed on individual computers. Provides personalised protection for each device.

Packet Filtering

Examines individual data packets and blocks them based on IP addresses and ports.

Stateful Inspection

Tracks active connections and makes decisions based on context and connection state.

Advantages of Firewalls

Enhanced Security

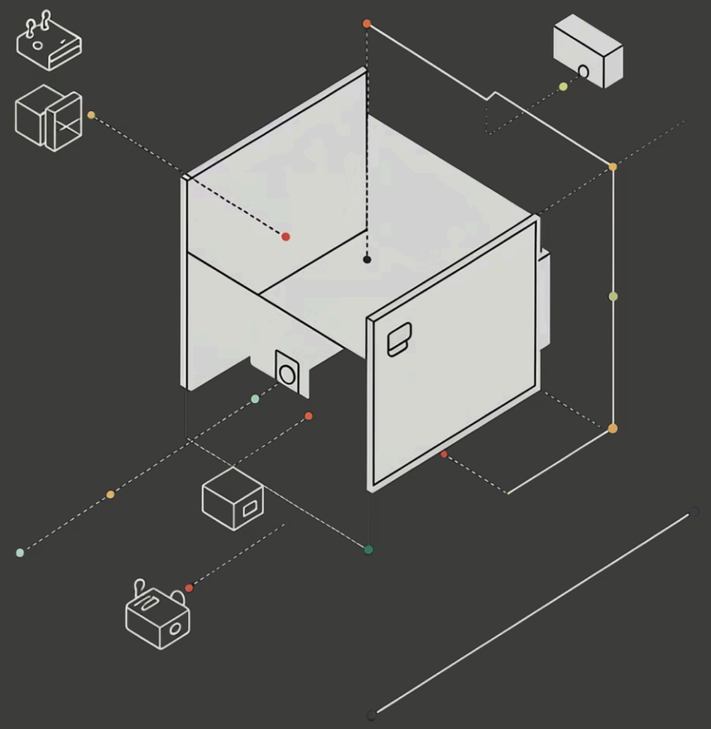
Blocks malicious software and unauthorised access attempts

Privacy Protection

Prevents hackers from accessing sensitive personal information

Traffic Monitoring

Provides visibility into network activity and potential threats



Disadvantages of Firewalls

Not Foolproof

Cannot stop all types of attacks, especially new or sophisticated malware that hasn't been identified yet.

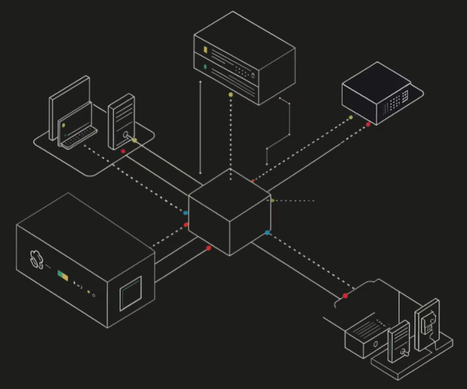
Performance Impact

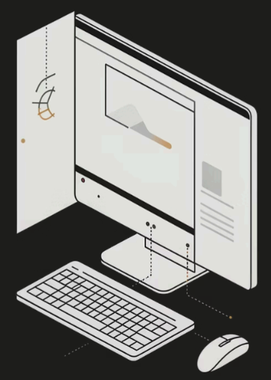
Can slow down network speed as it processes every packet, particularly with complex rule sets.

Configuration Challenges

Requires proper setup and regular updates to remain effective against evolving threats.

Real-life Examples





Corporate Networks

Large organisations use enterprise firewalls to protect thousands of computers, servers, and sensitive business data from external threats.

Personal Computers

Individual users benefit from software firewalls like Windows Defender, protecting against viruses and unauthorised access on home networks.

Essential for Modern Network Security